BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: PR–X (A-I-A4) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** PR–X (A-I-A4) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The arithmetic mean of the given data is: 2,5,10,11,12,15,15

(A) 10 (B) 11

(C) 12 (D) 15

Q2. The average of 58 and 88 with weights of 3 and 2 respectively is:

(A) 70 (B) 73

(C) 76 (D) 164

Q3. The area of sector of a circle whose radius is 4m, with central angle 3 radians is:

(A) 6 m^2 (B) 12 m^2

(C) 24 m^2 (D) 48 m^2

Q4. One of the coterminal angles of -120° is:

(A) -240° (B) -60°

(C) 120° (D) 240°

Q5. The length of two congruent chords which are both 8cm away from the centre of a circle of radius 10cm, is:

(A) 2 cm (B) 6 cm

(C) 12 cm (D) 18 cm

Q6. The joint variation in the given is expressed by:

(A) $q=3p^2$ (B) $p^2q=3$

(C) $q=3p^2r$ (D) $p^2qr=3$

Q7. The improper fraction in the following is:

(A) $(x^3+2)/(x^4+4)$ (B) $(x^3+4)/(x^4+2)$

(C) $(x^3+4)/(x^4+4)$ (D) $(x^2+2)/(x^3+2)$

Q8. If A={1,2,3} and B={3,4,5}, then B\A=____

(A) {3} (B) {1,2}

(C) {4,5} (D) {1,2,4,5}

Q9. The range of R={(a,b),(c,a),(d,c)} is:

(A) {a,b,c} (B) {a,b,d}

(C) {a,c,d} (D) {a,b,c,d}

Q10. If A={1,2,3} and B={4,5,6}, then which of the given binary relations is a function from B to A?

(A) R={(1,4),(2,5),(1,6)} (B) R={(1,4),(2,5),(3,4)}

(C) R={(4,1),(5,2),(4,3)} (D) R={(4,1),(5,2),(6,1)}

Q11. The quadratic equation in the given is:

(A) $7x^2-3x+8=0$ (B) $7x^3-3x^2+8=0$

(C) $7x^3-3x+8=0$ (D) $7x^4-3x+8=0$

Q12. To apply the quadratic formula to $3x^2+2x=5$, we have;

(A) a=3, b=2, c=5 (B) a=3, b=-2, c=5

(C) a=3, b=-2, c=-5 (D) a=3, b=2, c=-5

Q13. Discriminant of the quadratic equation $x^2-2x-5=0$ is:

(A) -24 (B) -16

(C) 16 (D) 24

Q14. Product of the roots of the quadratic equation, $3x^2-6x+7=0$ is:

(A) -7/3 (B) -2

(C) 2 (D) 7/3

Q15. The inverse variation in the given is expressed by:

(A) 4:5 (B) 4:5=8:10

(C) y=5x (D) xy=5

SECTION-B

Time Allowed: 2 Hours 40 Minutes

Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 40 minutes.

Q2. Answer any NINE parts. Each part carries FOUR marks.

i) Solve the given quadratic equation using factorization method. $x^2 + 2x - 3 = 0$

ii) Solve the given radical equation: $\sqrt{(2x-1)} = x-2$

iii) For what value of k, the roots of the equation $kx^2 + 3x + 2 = 0$ are real.

iv) Use synthetic division to find Quotient and remainder when polynomial $x^4+2x^2-6x-12$ is divided by $x+4$.

v) Find the mean proportion of p^2-q^2 and $(p-q)/(p+q)$

vi) Resolve into partial fraction: $(2x+1)/(x^2+4x-5)$

vii) If A={a,e,i,o,u} and B={b,c,d,e,f}, then verify commutative property of union.

viii) Find variance for the given data: 16,18,20,22,24.

ix) If length of an arc of circle is 9cm which subtends an angle of 60°, find the radius of the circle.

x) If $\tan\theta = -3/2$, find the trigonometric ratios when θ lies in 2nd quadrant.

xi) If the length of tangent drawn to the circle from a point is 8cm and radius of circle is 6cm, then find the distance from center of circle to that point.

xii) Prove that equal chords of a circle subtend equal angles at the centre.

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 16

Note: Note: Attempt any THREE questions. All questions carry equal marks.

Q3. In ΔPQR, mPQ = 6cm, mQR = 5cm and mPR = 12cm. If S is the midpoint of side mPR, find the length of median QS.

Q4. Prove that one and only one circle can pass through three non-collinear points.

Q5. Prove that the opposite angles of any quadrilateral inscribed in a circle are supplementary.

Q6. Inscribe a square in a circle of radius 2.5 cm.